

Here is the simple, step-by-step breakdown of how these parts connect to make one USB Type-C port do all three jobs (charging, data, and audio).

Think of the 217B-BE09 (the USB Type-C port) as the main door. We are going to wire different pins on this door to specific helper chips.

Step 1: Set Up Power (Charging)

First, we connect the power pins so the device can charge.

Port Pins VBUS (A4, B4, A9, B9) \rightarrow Connect these directly to your system's Battery Charger or Power Management IC.

Port Pins GND (A1, B1, A12, B12, A1, B1) \rightarrow Connect these to your main circuit board Ground.

Step 2: Detect What is Plugged In (TUSB322)

The port needs to know if you plugged in a charger, a computer, or headphones. The TUSB322 chip does this.

Port Pin CC1 (A5) \rightarrow Connect to TUSB322 CC1 pin.

Port Pin CC2 (B5) \rightarrow Connect to TUSB322 CC2 pin.

How it works: The TUSB322 checks these pins. If it sees analog headphones, it sends a signal to your main processor to turn on "Audio Mode."

Step 3: Fast Data Transfer (USB 3.0)

For high-speed USB 3.0 data, we bypass the switches entirely and go straight to the main brain (your processor).

Port Pins TX1/RX1 & TX2/RX2 (A2, A3, B10, B11, A10, A11, B2, B3) \rightarrow Connect these directly to the SuperSpeed USB 3.0 pins on your main processor.

Step 4: Share the D+/D- Pins for Audio & USB 2.0 (TS5USBA224)

The D+ and D- pins on a Type-C port do double-duty. They handle standard USB 2.0 data, but when headphones are plugged in, they carry the Left and Right audio. The TS5USBA224 is a switch that flips between the two.

Port Pin D+ (A6, B6) \rightarrow Connect to TS5USBA224 COM1 pin.

Port Pin D- (A7, B7) \rightarrow Connect to TS5USBA224 COM2 pin.

Switch Path A (Data): Connect the NC1 / NC2 pins on the switch to your processor's USB 2.0 pins.

Switch Path B (Music): Connect the NO1 / NO2 pins on the switch to the Left and Right audio outputs on your TLV320AIC3204 Audio Codec.

Step 5: Handle the Microphone (TS3A227E)

Headphones also have a microphone and a ground connection. These travel over the "SBU" pins on the Type-C port. The TS3A227E chip detects the microphone and routes it correctly.

Port Pin SBU1 (A8) \rightarrow Connect to TS3A227E RING2 / MIC pin.

Port Pin SBU2 (B8) \rightarrow Connect to TS3A227E SLEEVE / GND pin.

Microphone Output: Connect the MIC output pin on the TS3A227E directly to the microphone input on your TLV320AIC3204 Audio Codec.

Step 6: Process the Audio (TLV320AIC3204)

The TLV320AIC3204 is the audio brain. It takes digital music from your processor and turns it into real sound.

Digital side: Connect the I2S pins (BCLK, WCLK, DIN, DOUT) to your main processor so it can receive digital music.

Analog side: Its Left/Right outputs go to the switch in Step 4. Its microphone input comes from the switch in Step 5.

In Summary: Power goes straight to the charger. High-speed data goes straight to the processor. The CC chip (TUSB322) acts as the traffic cop, telling the audio switches (TS5USBA224 and TS3A227E) whether to let USB data pass through, or to route music and microphone signals to the audio chip (TLV320AIC3204).

It is completely understandable to want a clear, top-to-bottom overview. Designing a circuit with multiple chips can feel like trying to solve a complex puzzle, but it all makes sense once you see how they work together as a team.

Let us walk through the complete, end-to-end journey of how your device handles charging, data, and audio through a single port.

Step 1: The Physical Entry Point (217B-BE09)

Everything starts at the **217B-BE09**. This is simply the physical metal USB Type-C port on the outside of your device. It has 24 tiny metal pins inside. When you plug in a cable, those pins connect with the cable's pins. Every single signal—whether it is electricity from a wall charger, a movie from a flash drive, or music to your headphones—must enter and exit through this physical gateway.

Step 2: The Identification Process (TUSB322)

The moment you plug something into the port, the device needs to know what it is. It cannot send music to a battery charger, and it cannot pull power from a pair of headphones.

This is where the **TUSB322** steps in. Think of it as the security guard at the front desk.

- It monitors two specific pins on the USB port called the Configuration Channel (CC).
- By checking these pins, the TUSB322 instantly identifies what you just plugged in (a charger, a computer, or an analog audio headset).
- Once it knows what is connected, it sends electronic signals to the rest of the chips on your circuit board to tell them how to behave.

Step 3: Handling Power and Fast Data (Direct Routes)

Before we talk about the complicated switching, let us handle the easy stuff:

- **Charging:** If the TUSB322 detects a charger, the power flows directly from the port's power pins (VBUS) straight to your battery charger chip. The other chips ignore it.
- **Fast USB 3.0 Data:** If you plug in a hard drive, the high-speed data uses its own dedicated "express lane" pins (TX and RX). This data flows directly to your device's main computer brain (the processor).

Step 4: The Audio vs. Data Crossroads (TS5USBA224)

Here is where the design gets clever. A USB-C port does not have enough pins to give both standard USB 2.0 data and analog audio their own dedicated paths. They have to share a pair of pins (called D+ and D-).

The **TS5USBA224** is a physical, electronic track-switcher.

- If you plug in a normal USB cable (like connecting to a PC), the security guard (TUSB322) tells this switch to route the D+ and D- pins to your device's data processor.
- If you plug in an analog headphone adapter, the security guard tells this switch to flip. It disconnects the data processor and instead connects the port directly to your audio system. Now, those same pins carry Left and Right music signals.

Step 5: The Microphone Detective (TS3A227E)

If you are using a headset, you probably have a built-in microphone to make calls. Headphones have a few extra pins for this (called SBU pins).

However, different brands of headphones wire their microphones and ground wires differently. The **TS3A227E** chip is an automatic detector. It listens to those SBU pins, figures out exactly how your specific headphones are wired, and automatically connects the microphone signal to the right place so your voice comes through clearly.

Step 6: The Sound Translator (TLV320AIC3204)

Finally, we reach the heart of the audio system. The **TLV320AIC3204** is an Audio Codec. Its job is translation.

- **Playing Music:** Your phone or computer only understands digital 1s and 0s. The Codec takes that digital code and translates it into physical, analog electrical waves. Those waves travel through the switch (Step 4), out the USB port (Step 1), and physically move the speakers in your headphones so you hear music.
- **Recording Voice:** When you speak, the Codec takes the analog electrical waves coming from your microphone (Step 5) and translates them back into digital 1s and 0s so your computer can process your voice.

End-to-End Summary

When you plug a cable in, the physical port (**217B-BE09**) makes the connection. The detector (**TUSB322**) figures out what you plugged in. If it is headphones, it triggers the main switch (**TS5USBA224**) to allow music to flow, while the mic manager (**TS3A227E**) ensures your microphone is connected properly. Finally, the audio brain (**TLV320AIC3204**) does the actual work of turning digital files into beautiful sound and vice versa.